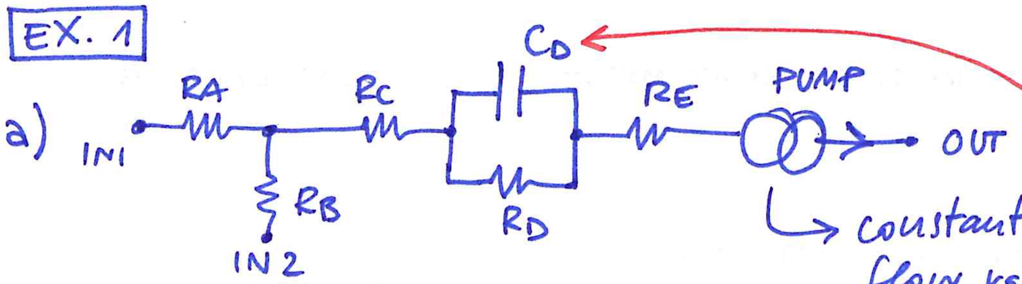
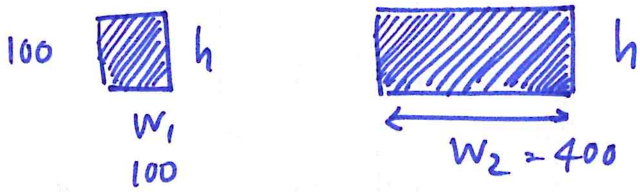
EX. 1

a)



ROOF FLEXIBLE $\downarrow$ CAPACITANCE

$\hookrightarrow$ constant flow rate $\longrightarrow$ CURRENT SOURCE



$h = 100 \longrightarrow R_D = \dfrac{12 \cdot \eta L}{\left(1 - 0.63\frac{h}{w}\right) h^3 \cdot w} = \boxed{1.78 \cdot 10^{11}}$ $\left[\dfrac{Pa \cdot s}{m^3}\right]$

$\hookrightarrow r_{eq} = 50\,\mu m \qquad R = \dfrac{8 \cdot \eta L}{\pi\, r_{eq}^4} \qquad \gamma = \dfrac{8 \cdot \eta}{\pi \cdot r_{eq}^4} = 4 \cdot 10^{14}$

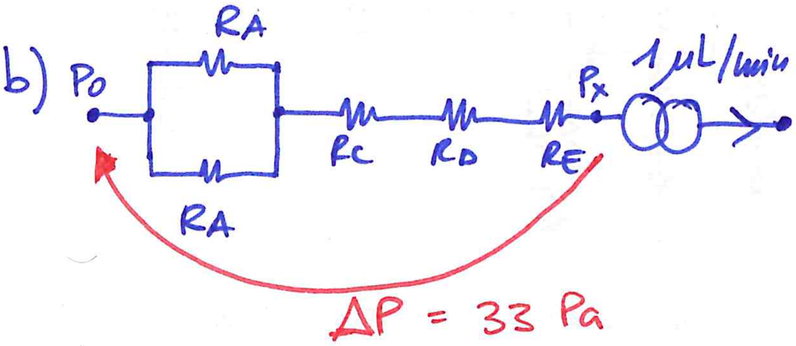
$R_A = \gamma \cdot 10^{-3} = \boxed{4 \cdot 10^{11} \left[\dfrac{Pa \cdot s}{m^3}\right]}$

$R_B = R_A$

$R_C = 2 \cdot R_A = \boxed{8 \cdot 10^{11} \left[\dfrac{Pa \cdot s}{m^3}\right]}$

$R_E = R_C$

b)



$V_E = V_C = \dfrac{Q}{h \cdot W_1} = \dfrac{10^{-6} \frac{L}{m}}{100 \cdot 400} = \boxed{1{,}7\, mm/s}$

$V_A = \dfrac{Q/2}{h \cdot W_1} = \dfrac{V_E}{2} = \boxed{0.83\, mm/s}$

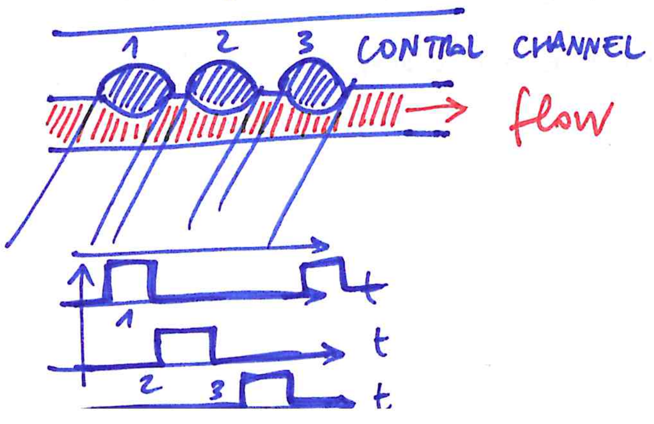
$V_B = \dfrac{Q}{h \cdot W_2} = \dfrac{V_E}{4} = \boxed{0.41\, \frac{mm}{s}}$

$P_x = P_0 - \Delta P =$

$= P_0 - \underbrace{Q \cdot \left(\frac{R_A}{2} + R_C + R_D + R_E\right)}_{33\,Pa} = 101 \cdot 000\ Pa - 33\ Pa =$

$= \boxed{100 \cdot 967\ Pa}$

c) 2-layer structure



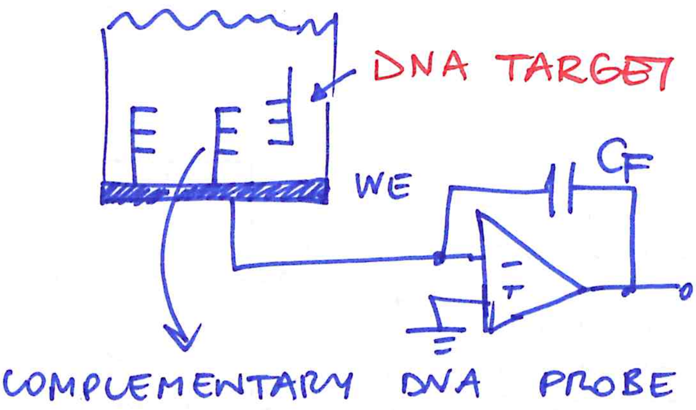
CROSS-BAR TECHNOLOGY

PRO: COMPACT
SIMPLE
NO CONTACT WITH FLUID
INFINITE PUMPING VOLUME

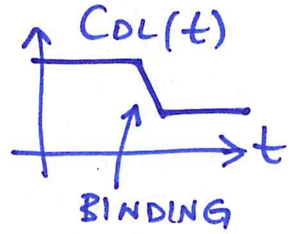
CON: _ PULSATILE FLOW

EX. 2

a)

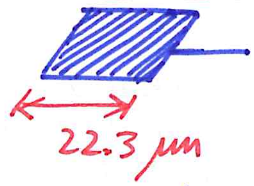


$\rightarrow$ REMOVE IONS $\downarrow$ DECREASE C_{DL}

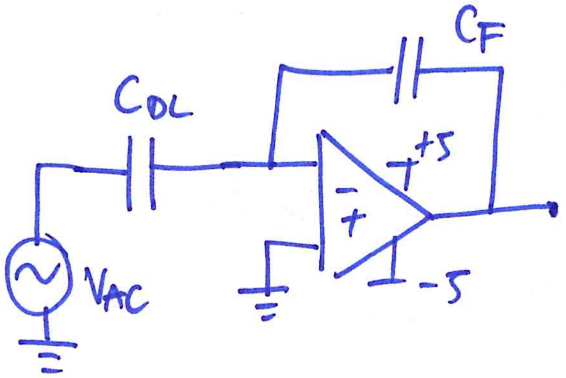


b) $C = C_{DL} = C_0 \cdot A = \frac{0.1\,pF}{\mu m^2} \cdot A = 50\,pF \rightarrow \boxed{A = 500\,\mu m^2}$

$C_0 \rightarrow$ IDEAL



c)



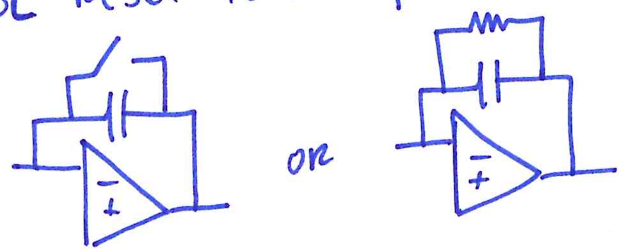
I set $V_{AC} = 10\,mV$ to be SMALL

$|V_{OUT}| = \left| -V_{AC} \cdot \frac{C_{DL}}{C_F} \right| = 5\,V$ max

$C_F = C_{DL} \cdot \frac{10\,mV}{5\,V} = 50\,pF \cdot \frac{10}{5000} = 0.1\,pF$

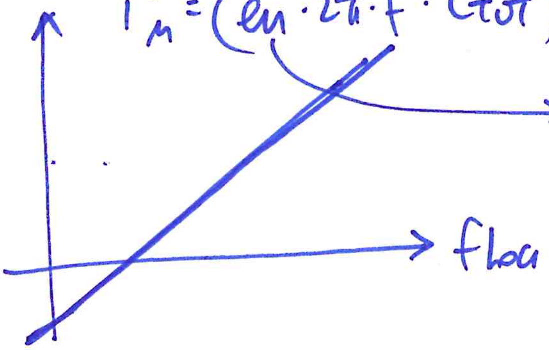
$\rightarrow$ [VERY SMALL !]

d) DC RESET is missing



e) $i_n^2 = (e_n \cdot 2\pi \cdot f \cdot C_{TOT})^2$ $\quad C_{TOT} = C_{DL} + C_F = 100\,pF$

$e_n \rightarrow$ from plot (neglect $1/f$)



$4\,nV/\sqrt{Hz}$

f)